

JUN YAMAMOTO

PHD CANDIDATE IN NETWORK SCIENCE

Department of Network and Data Science
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RESEARCH PROFILE

I research organizational principles of network structure and dynamics using statistical physics. My current work focuses on physical networks, network coarse-graining and scale-invariance, and network diffusion and synchronization.

EDUCATION

- 2023–2027 (expected) **PhD Network Science**, Central European University, Vienna, Austria
Supervisors: Prof. Dr. Márton Pósfai and Prof. Dr. János Kertész
Thesis: “Dynamical processes on physical networks” (tentative).
- 2026/06–07 AccelNet-MultiNet Fellowship Research Visit, Center for Complex Networks Research, Department of Physics, Northeastern University, Boston, USA
Host PI: Prof. Dr. Albert-László Barabási
- 2022–2023 **MSc Mathematics**, Queen Mary University of London, London, UK
Supervisor: Prof. Dr. Ginestra Bianconi
Thesis: “Modelling higher-order network dynamics in the presence of triadic interactions.”
- 2017–2022 **BEng Applied Physics**, Hokkaido University, Sapporo, Japan
Supervisor: Prof. Dr. Kousuke Yakubo
Thesis: “Bifractality of fractal scale-free networks.”
- 2019–2020 Exchange Studies, Department of Physics, ETH Zürich, Zürich, Switzerland

PUBLICATIONS

1. M. Niedostatek, A. Baptista, **J. Yamamoto**, J. Kurths, R. Sanchez Garcia, B. MacArthur, and G. Bianconi, “Mining higher-order triadic interactions.” *Nature Communications* **16**, 11613 (2025). DOI: 10.1038/s41467-025-66577-z
2. K. Yakubo, G. Shimojo, and **J. Yamamoto**, “Random walks on bifractal networks.” *Physical Review E* **110**, 064318 (2024). DOI: 10.1103/PhysRevE.110.064318
3. **J. Yamamoto** and K. Yakubo, “Bifractality of fractal scale-free networks.” *Physical Review E* **108**, 024302 (2023). DOI: 10.1103/PhysRevE.108.024302

PREPRINTS

1. **J. Yamamoto**, I. Bonamassa, and M. Pósfai. “Localization transitions of diffusion dynamics in physical networks”, arXiv:2607.19486 (2026).
2. **J. Yamamoto**, I. Bonamassa, and M. Pósfai. “Laplacian eigenmode localization in physical complex networks”, manuscript submitted (2026). Joint submission with arXiv:2607.19486.

CONFERENCE PRESENTATIONS

- 2025 **J. Yamamoto**, I. Bonamassa, and M. Pósfai. *Localization-delocalization transition in physical networks*. The 29th International Conference on Statistical Physics (StatPhys29), Florence, Italy. July 14, 2025.
- J. Yamamoto**, I. Bonamassa, and M. Pósfai. *Delocalization of physical Laplacian eigenvectors due to degree-volume correlation*. The International School and Conference on Network Science (NetSci2025), Maastricht, the Netherlands. June 4, 2025.
- 2022 **J. Yamamoto** and K. Yakubo. *Bifractal property of stochastic scale-free networks*. The Physical Society of Japan 77th Annual Meeting, online. March 15, 2022.
- 2021 **J. Yamamoto** and K. Yakubo. *Multifractality of scale-free networks*. The Physical Society of Japan 76th Annual Meeting, online. September 20, 2021.

CONFERENCE POSTERS

- 2026 **J. Yamamoto**, I. Bonamassa, and M. Pósfai, *When node weight matters: spectral transitions in physical networks*. Network Science Seminar 2025, Kyoto, Japan, February 17–20, 2026.
- 2022 **J. Yamamoto**, G. Shimojo, and K. Yakubo, *Bifractal nature of fractal scale-free networks and its implications*. Network Science Seminar 2022, Kyoto, Japan, August 23–25, 2022.
- 2021 **J. Yamamoto** and K. Yakubo, *Bifractality of scale-free networks*. Network Science Seminar 2021, Kanazawa, Japan, December 11–12, 2021. *Poster Award*.

SEMINARS AND WORKSHOPS

- 2025 **J. Yamamoto**, *Physical Networks as Network-of-Networks*. Network Science Study Group (ネットワーク科学勉強会), Online. May 25, 2025.
- J. Yamamoto**, I. Bonamassa, and M. Pósfai, *Reentrant localization in physical Laplacian eigenvectors*. DYNASNET workshop on graphs, networks, and brains, Budapest, Hungary. May 13, 2025.
- 2024 **J. Yamamoto**, I. Bonamassa, and M. Pósfai. *Laplacian Spectra and Angular Constraints in Physical Networks*. Annual DYNASNET Meeting, Valtice, Czech Republic. October 16, 2024.

FELLOWSHIPS AND GRANTS

- 2026 AccelNet-MultiNet Fellowship, National Science Foundation. Research visit to the Center for Complex Networks Research, Northeastern University, Boston, USA, to work with Prof. Dr. A.-L. Barabási. (\$4,000).
- 2022–2023 ITO Scholarship for Japanese Students, ITO Foundation. Tuition support, monthly stipend, and travel allowance for master’s study in the UK. (~\$50,000).

SELECTED AWARDS AND HONORS

2025	First-Year PhD Award, Central European University.
2023	Principal's Prize, Queen Mary University of London.
2022	Dean's Award for Academic Achievement, Hokkaido University.
2021	Poster Award, Network Science Seminar 2021, Kanazawa, Japan.

TEACHING

2024–2026	Teaching Assistant, Central European University. Scientific Python (graduate); Introduction to Linear Algebra (undergraduate).
2021–2023	Teaching Assistant, Hokkaido University. Statistical Mechanics I; Applied Mathematics II; Computational Science; Introduction to Statistics.

ACADEMIC SERVICE

2025– present	Ad hoc reviewer, <i>Nature Communications</i> ; <i>Journal of Complex Networks</i> ; <i>Transactions of the Japanese Society for Artificial Intelligence</i> .
2025–2026	Organizer, Physical Networks IV Satellite, NetSci2026, Boston, USA.
2024–2025	Organizer, Physical Networks III Satellite, NetSci2025, Maastricht, the Netherlands.
2024–2025	Organizer, Data Stories 11th Edition, Department of Network and Data Science, Central European University.

OTHER EXPERIENCE

2022	Data Scientist, Fujitsu, DX Large Data Platform Business Group. Natural language processing and network analysis for large-scale Japanese text data.
2020	Summer Research Intern, Japan Atomic Energy Agency. Machine learning algorithms for accelerated molecular dynamics simulation of materials.
2018	Research Intern, Okinawa Institute of Science and Technology. Electron microscopy and vacuum system development for protein crystallography.

SKILLS

Programming Python, C++, Julia, Fortran.

Tools Linux, Git, L^AT_EX, Mathematica.

Languages Japanese (native), English (fluent), German (beginner).

REFEREES

Prof. Dr. Márton Pósfai, Department of Network and Data Science, Central European University, Vienna, Austria. *PhD supervisor*.

Prof. Dr. Ginestra Bianconi, School of Mathematical Sciences, Queen Mary University of London, London, UK. *MSc supervisor*.

Prof. Dr. Kousuke Yakubo, National Institute of Technology, Asahikawa College, Asahikawa, Japan. *Undergraduate supervisor*.

Prof. Dr. Ivan Bonamassa, International Research Center for Complexity Science, Hangzhou International Innovation Institute (H3I), Beihang University, Hangzhou, China. *Research collaborator*.